List of publications

Jakub Bajzík

1. Bajzik J, Depope A, Zolfimoselo Y, Sharipov A, Lesayova A, Klein H, Richmond A, Vernardis S, Grauslys A, Andrejev S, Zelezniak A, Ralser M, Marioni RE, Mondelli M, Robinson MR. Joint Variable Selection for Omic Biomarkers in Time-to-Event Data. *bioRxiv*. 2026. doi:10.64898/2026.04.30.721585.

2. Kögler K, Bajzik J, Robinson MR, Mondelli M. I Can't Believe It's the Encoding: Better than Deep Learning via an Interpretable Linear Model in Proteomic Prediction. In: *I Cant Believe Its Not Better (ICBINB): Failure Modes of AI in Biology*. NeurIPS. 2026. URL: https://openreview.net/forum?id=vzfbyRXaXh.

3. Robertson JA, Krätschmer I, Richmond A, McCartney DL, Bajzik J, Vernardis S, Corley J, Tomlinson SJ, Vieno M, Chybowska AD, Grauslys A, Smith HM, Brigden C, Messner CB, Zelezniak A, Ralser M, Russ TC, Pearce J, Cox SR, Robinson MR, et al. Air pollution exposure in Generation Scotland: molecular fingerprints and health outcomes. *medRxiv*. 2026. doi:10.64898/2026.03.04.26347573.

4. Bajzik J, Depope A, Zolfimoselo Y, Sharipov A, Klein H, Mondelli M, Robinson MR. Joint Variable Selection in Proteomics Survival Models. In: *ICLR 2026 Workshop on Machine Learning for Genomics Explorations*. 2026. URL: https://openreview.net/forum?id=Re1rA204zd.

5. Depope A, Bajzik J, Mondelli M, Robinson MR. Joint modeling of whole-genome sequencing data for human height via approximate message passing. *Cell Genomics*. 2026;6(5):101162. doi:10.1016/j.xgen.2026.101162.

6. Robertson JA, Bajzik J, Vernardis S, Chybowska AD, McCartney DL, Grauslys A, Mur J, Smith HM, Campbell A, Drake C, Grant H, Pearce J, Russ TC, Adkin P, White M, Brigden C, Messner CB, Porteous DJ, Hayward C, Cox SR, et al. Methylome-wide association studies and epigenetic biomarker development for 133 mass spectrometry-assessed circulating proteins in 14,671 Generation Scotland participants. *Genome Biology*. 2025;26(1):417. doi:10.1186/s13059-025-03892-0.

7. Bajzik J, Prinosil J, Jarina R, Mekyska J. Independent Channel Residual Convolutional Network for Gunshot Detection. *International Journal of Advanced Computer Science and Applications*. 2022;13(4). doi:10.14569/IJACSA.2022.01304108.

8. Bajzik J, Jarina R. Sound event detection using class activation maps. In: *2022 ELEKTRO (ELEKTRO)*. 2022, 1–5. doi:10.1109/ELEKTRO53996.2022.9803350.

9. Bajzik J, Jarina R. Exploiting hierarchy in environmental sound classification. In: *2022 32nd International Conference Radioelektronika (RADIOELEKTRONIKA)*. 2022, 1–4. doi:10.1109/RADIOELEKTRONIKA54537.2022.9764900.

10. Volak J, Bajzik J, Janisova S, Koniar D, Hargas L. Real-Time Interference Artifacts Suppression in Array of ToF Sensors. *Sensors*. 2020;20(13). doi:10.3390/s20133701.

11. Bajzik J, Prinosil J, Koniar D. Gunshot Detection Using Convolutional Neural Networks. In: *2020 24th International Conference Electronics*. 2020, 1–5. doi:10.1109/IEEECONF49502.2020.9141621.

12. Bajzik J, Koniar D, Hargas L, Volak J, Janisova S. Depth Sensor Selection for Specific Application. In: *2020 ELEKTRO*. 2020, 1–6. doi:10.1109/ELEKTRO49696.2020.9130293.

13. Koniar D, Volak J, Hargas L, Janisova S, Bajzik J. Depth cameras calibration for obstructive sleep apnea diagnostic support. In: *2020 ELEKTRO*. 2020, 1–6. doi:10.1109/ELEKTRO49696.2020.9130320.

14. Bajzik J. DEEP LEARNING BASED SOUND EVENT RECOGNITION. In: *Proceedings of the 25thConference STUDENT EEICT 2019*. STUDENT EEICT 2019. 2019, 382–385.